Risk-Controlled Spatio-Temporal Graph Learning for Anti-Money Laundering Under Extreme Imbalance and Topological Camouflage

Md. Nazmul, Musrat Jahan Gungun, and Maheli Ahmed

***Abstract*—Financial money laundering moves an estimated \$800B–\$2T annually. While Graph Neural Networks (GNNs) capture relational transaction topologies, real-world Anti-Money Laundering (AML) deployment is impeded by velocity evasion, topological camouflage, extreme class imbalance ($\pi < 0.05\%$), cold-start isolation, and uncalibrated decision uncertainty. We propose C-STGB (Conformal Spatio-Temporal GraphBoost), a unified geometric deep learning framework integrating continuous tri-band harmonic temporal attention, learnable edge-trust gating, typology-clustered latent GraphSMOTE, evidence-adaptive tabular fusion, cost-sensitive Bayes risk optimization, and Class-Conditional Conformal Risk Control (CRC). Benchmarked across 14 financial networks (9.53M entities, 9.53M transactions) across 182 dataset-model pairs (14 datasets × 13 models, 12 baselines plus C-STGB, over five seeds = 910 executions), C-STGB achieves top-tier performance, reaching 99.44% F1 on Elliptic-v1 under invariant-augmented Bayes calibration (PR-AUC 1.0000), 100.00% F1 on Elliptic-v2, and 99.80% on PaySim Extended, yielding an overall 14-dataset unweighted macro F1 of 67.26% (0.6848 PR-AUC; competitive with tuned GBDTs at 67.92% which excel on flat single-hop data, while delivering decisive gains on multi-hop topologies and achieving statistically significant superiority over deep spatial and dynamic GNN baselines with a +43.49 pp mean uplift over GCN; $W = 105.0, W_- = 0.0, n = 14, p = 1.22 \times 10^{-4}, p_{\text{adj}} < 0.001$ under two-sided Wilcoxon signed-rank tests with Benjamini–Hochberg FDR correction). Furthermore, C-STGB provides distribution-free class-conditional coverage ($\geq$ 99.0%) under within-class exchangeability, routes > 99.4% of volume through straight-through execution tiers, and achieves an end-to-end streaming Fast-Path inference latency of 0.45 ms per single event (0.12 ms GPU forward pass, and 0.054 ms amortized forward pass under batch-64 inference) on benchmark hardware.**

***Index Terms*—Anti-Money Laundering, Graph Neural Networks, Temporal Graph Learning, Conformal Prediction, Financial Forensics.**

I. Introduction

FINANCIAL crime surveillance is a critical societal application of machine learning in global financial infrastructure. According to the United Nations Office on Drugs and Crime (UNODC) and Financial Action Task Force (FATF), illicit syndicates launder between \$800 billion and \$2 trillion annually (2% to 5% of global GDP) across banking rails, mobile financial service (MFS) e-wallets, and crypto-asset ledgers [1], [2]. Traditional Financial Intelligence Units (FIUs) rely heavily on static heuristics triggering excessive false positive alerts (>90% to 95%), placing a heavy operational burden on human compliance teams.

While Graph Neural Networks (GNNs) capture relational topologies [3]–[7], practical AML deployment faces five foundational technical barriers: (1) **Continuous-Time Velocity Evasion:** microsecond bursts vs. 90-day dormant holding chains cannot be captured simultaneously by discrete snapshots (e.g., EvolveGCN [8]) or single exponential decays; (2) **Adversarial Topological Camouflage:** connecting illicit flows to high-degree commercial hubs induces severe feature oversmoothing in standard GCN/GAT aggregation [9]; (3) **The Bayes Risk Imbalance Barrier ($\pi < 0.05\%$):** severe class skew causes posterior attenuation, yielding recall collapse under standard thresholds ($\tau = 0.50$); (4) **Cold-Start Graph Isolation ($d(u) = 0$):** virgin accounts receive zero-padded message vectors in spatial aggregation; and (5) **Uncalibrated Decision Uncertainty:** black-box scores lack statistical validity and safety guarantees under temporal distribution drift.

To resolve these challenges, we formulate and test three core hypotheses: (H_1) continuous multi-scale temporal attention with learnable edge-filtering suppresses over-smoothing under burst and dormancy dynamics, improving minority PR-AUC relative to static and single-decay dynamic GNNs; (H_2) interpolating minority representations within latent typology clusters (C_k) recovers minority recall without increasing false-positive rates on benign hubs under extreme class imbalance; and (H_3) Class-Conditional Conformal Risk Control (CRC) delivers finite-sample coverage ($\geq$ 99%) under within-class exchangeability while routing the vast majority (> 99%) of transaction volume through decisive straight-through execution tiers.

We present **C-STGB** (Conformal Spatio-Temporal GraphBoost), a unified geometric deep learning framework for robust anti-money laundering detection. Our primary contributions span:

1) *Continuous-Time Multi-Scale Attention:* Integrates logarithmic harmonic temporal encodings (Φ_{time}) with Hawkes arrival velocity gates and learnable edge-trust filtering ($\mathbf{g}_{ij}$), attenuating spurious camouflage edges during message passing.
2) *Typology-Clustered Latent GraphSMOTE:* Constrains latent oversampling along empirical laundering clusters (C_k)

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with bilinear link generation and Asymmetric Focal Tversky loss, raising minority recall to 98.88% under domain-informed Bayes calibration (99.44% F1 on Elliptic-v1).

3) *Class-Conditional Conformal Risk Control:* Establishes distribution-free finite-sample coverage ($\geq$ 99.0%) under within-class exchangeability, complete with non-asymptotic total variation drift bounds and streaming Adaptive Conformal Inference (ACI).
4) *Comprehensive 14-Dataset Benchmark Corpus:* Evaluates 9.53M entities and 9.53M transactions across 182 dataset-model configuration pairs (14 datasets $\times$ 13 models, 12 competitive baselines plus C-STGB, evaluated over five random seeds = 910 experimental executions). C-STGB achieves competitive classification across diverse topologies (99.44% on Elliptic-v1, 100.00% on Elliptic-v2, 99.80% on PaySim Extended; 67.26% 14-dataset unweighted macro F1) and statistically significant superiority over deep spatial and dynamic GNN baselines (+43.49 pp over GCN, $W = 105.0, W_{-} = 0.0, n = 14, p_{\text{adj}} < 0.001$), while performing competitively with optimized gradient boosted trees (CatBoost 67.14%, XGBoost 67.92%) and delivering marked gains on complex multi-hop relational networks.
5) *Low-Latency Triage & Assisted Decision Support:* Achieves streaming Fast-Path inference latency of 0.45 ms per event (0.12 ms GPU forward pass, 0.054 ms batch-64 amortized pass), routing > 99.4% of volume through straight-through tiers, paired with an auxiliary compliance decision-support prototype for structured SAR drafting in Section VI-C.

Architectural Scoping: Primary learning is formulated as inductive entity classification over continuous transaction graphs, with transactions represented as relational edges or event vertices. Extensions such as lifelong learning across sequential ledgers and federated graph learning are delineated as modular future directions. The remainder of this paper is organized as follows: Section II reviews related work. Section III establishes the methodology. Section IV details empirical evaluations. Section V discusses operational implications. Section VI provides XAI case studies and SAR drafting. Section VII addresses limitations, and Section VIII concludes.

II. Related Work

A. Graph Representation Learning for Anti-Money Laundering

The application of Graph Neural Networks (GNNs) to financial forensic analysis was pioneered by Weber *et al.* [7] on the Elliptic Bitcoin dataset, demonstrating that spatial architectures like Graph Convolutional Networks (GCN) [5] and GraphSAGE [6] capture relational neighborhood context beyond isolated tabular classifiers. Subsequent extensions by Bellei *et al.* [10] introduced the Elliptic2 multi-asset subgraph dataset, emphasizing multi-hop subgraphs, while Johannessen and Jullum [11] explored heterogeneous graph representations across account ledgers. Beyond Bitcoin, graph embeddings have been investigated for smart contract fraud on Ethereum ledgers by Wu *et al.* [12] and Zheng *et al.* [13], suspicious exchange activity by Gandal *et al.* [14], multi-bank synthetic wire simulations by Lorenz *et al.* [15] and Suzumura *et al.* [16], streaming fraud detection by Dal Pozzolo *et al.* [17], and cross-institutional AML architectures by Liu *et al.* [18]. Recent work investigated inductive temporal representations [19] and catastrophic forgetting mitigation via Topology-Aware Weight Preservation (TWP) [20].

Remaining Research Gap: Standard spatial and heterogeneous GNNs assume static or homophilic connectivity and lack learnable edge-trust filtering. Consequently, when adversaries deliberately inject high-volume transaction chaff to connect with high-degree commercial hubs, standard message aggregation suffers from feature over-smoothing, diluting localized laundering signatures [9].

B. Continuous-Time Dynamic Temporal Graphs & Point Processes

Financial transactions are inherently continuous-time relational events. Discrete snapshot dynamic GNNs, such as EvolveGCN [8], discretize streams into time slices, destroying intra-slice ordering and obscuring sub-second bursts. While continuous frameworks like Temporal Graph Networks (TGN) [26], TGAT [19], and dynamic anomaly models [28] preserve event order, single exponential decay kernels ($\exp(-\lambda\Delta t)$) rapidly attenuate to zero over multi-week dormant periods. Point processes, formalized by Hawkes [29], model transaction clustering and velocity acceleration, while weight preservation counters non-stationary streaming drift [20].

Remaining Research Gap: Existing temporal models enforce uniform decay rates across all frequencies, failing to simultaneously capture microsecond velocity bursts alongside extended dormant holding intervals.

C. Imbalanced Graph Learning & Latent Oversampling

Financial transaction graphs exhibit extreme class imbalance ($\pi < 0.05\%$). Classical Euclidean oversampling methods (e.g., SMOTE [30]) lack topology awareness, generating synthetic instances without valid edges. Under severe skew, Precision-Recall (PR-AUC) analysis is essential for rigorous benchmarking [31]. To bridge topological oversampling, Zhao *et al.* [27] proposed GraphSMOTE, performing feature interpolation in the intermediate embedding space.

Remaining Research Gap: Conventional GraphSMOTE executes unconstrained latent interpolation across the global minority class pool, disregarding distinct structural typologies (cyclic wash trading vs. smurfing fan-out vs. pass-through layering). Unconstrained interpolation distorts domain flow regularity ($\Phi_{\text{flow}} \approx 1.0$), inducing false alarms on benign commercial accounts.

D. Finite-Sample Conformal Risk Control & Distribution Shift

Conformal prediction, formalized by Vovk *et al.* [32] and extended by Angelopoulos and Bates [33], [34], provides distribution-free, finite-sample uncertainty quantification. Recent advances investigated conformal prediction on GNNs [35] and class-conditional conformal prediction under class imbalance [36], [37]. While marginal conformal prediction guarantees valid overall error coverage $(1-\alpha)$, it suffers from

TABLE I

SYSTEMATIC METHODOLOGICAL COMPARISON: CAPABILITIES OF REPRESENTATIVE FRAUD AND AML ARCHITECTURES VS. PROPOSED C-STGB.

Model / Paradigm Architecture	**Continuous Multi-Scale Temporal Dynamics**	**Anti-Camouflage Edge-Trust Gating**	**Latent Typology Graph Oversampling**	**Cold-Start Graph-Tabular Fusion**	**Distribution-Free Conformal Risk Control**	**Benchmarked Single Inference Latency**
XGBoost / CatBoost / LightGBM [21]–[23]	× (Tabular only)	× (No Graph)	× (Euclidean SMOTE)	× (No Graph)	× (Heuristic τ)	✓ (< 0.20 ms)
GCN / GAT / GIN [5], [24], [25]	× (Static snapshot)	× (Uniform/Homophilic)	× (None)	× (Zero-padded)	× (Softmax score)	× (> 12.5 ms)
EvolveGCN [8]	× (Discrete slices)	× (No Edge Filter)	× (None)	× (Isolated nodes)	× (Uncalibrated)	× (> 45.0 ms)
TGN / TGAT [19], [26]	✓ (Single $\exp(-\lambda\Delta t)$)	× (No Camouflage Filter)	× (None)	× (Node memory)	× (Uncalibrated)	× (> 8.5 ms)
CARE-GNN (Adversarial) [9]	× (Static graph)	✓ (RL Edge Selection)	× (None)	× (No Tabular Fusion)	× (Uncalibrated)	× (> 18.0 ms)
GraphSMOTE [27]	× (Static graph)	× (No Filter)	✓ (Unconstrained Global)	× (No Fusion)	× (Uncalibrated)	× (> 14.2 ms)
Johannessen & Jullum [11]	× (Heterogeneous Static)	× (Static Aggregation)	× (None)	× (Feature Concat)	× (Uncalibrated)	× (> 16.5 ms)
Wang *et al.* (ACM KDD, 2024) [28]	✓ (Continuous Multi-Scale)	× (No Edge Gating)	× (None)	× (No Invariant Fallback)	× (Uncalibrated)	× (> 5.8 ms)
C-STGB (Proposed Framework)	✓ **(Tri-Band Harmonic)**	✓ **(Learnable MLP $\hat{g}_{ij}$)**	✓ **(Typology Manifold C_k)**	✓ **(Evidence-Adaptive α_u)**	✓ **(Finite-Sample ≥ 99%)**	✓ **(0.45 ms Fast-Path)**

severe conditional under-coverage on skewed datasets. Under temporal shifts, standard exchangeability must be explicitly bounded [38]–[40].

Remaining Research Gap & Contribution Differentiation: While Ding *et al.* [36] and Romano *et al.* [37] established class-conditional validity for static tabular classification, and Huang *et al.* [35] addressed static transductive node classification, existing methods do not provide finite-sample class-conditional coverage guarantees tailored for continuous-time financial graphs subject to temporal drift, nor do they operationalize prediction sets into auditable straight-through triage tiers. In this work, we formulate Class-Conditional Conformal Risk Control (CRC) tailored for continuous transaction networks, establishing finite-sample valid coverage per class under within-class exchangeability with explicit total variation drift tracking via streaming Adaptive Conformal Inference (ACI).

III. PROBLEM FORMULATION & C-STGB METHODOLOGY

A. Problem Formulation and Operational Objectives

We formulate financial crime surveillance over an evolving continuous-time transaction graph $\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t, \mathbf{X}_v^{(t)}, \mathbf{X}_e^{(t)})$, where $\mathcal{V}_t$ represents active financial accounts (entities), $\mathcal{E}_t = \{(u, v, t_e, \mathbf{e}_{uv}) : t_e \leq t\}$ denotes the chronological stream of directed fund transfers, $\mathbf{X}_v^{(t)} \in \mathbb{R}^{|\mathcal{V}_t| \times d_v}$ comprises vertex attributes, and $\mathbf{X}_e^{(t)} \in \mathbb{R}^{|\mathcal{E}_t| \times d_e}$ denotes transaction edge features (amount, currency rail, fee). Each entity $u \in \mathcal{V}_t$ carries a ground-truth regulatory classification label $y_u \in \{0, 1\}$, where $y_u = 1$ denotes an illicit entity and $y_u = 0$ denotes a verified benign entity, under extreme class prevalence skew $\pi = \mathbb{P}(y_u = 1) \ll 0.01$. In transaction streams lacking explicit account identity graphs (e.g., PaySim), transactions are mapped to dedicated transaction-event vertices, establishing an exact 1:1 entity-to-transaction formulation.

The operational surveillance task encompasses two complementary objectives:

1) **Entity Classification Objective:** Learn an inductive hypothesis $f_\Theta : \mathcal{V}_t \times \mathcal{G}_t \to [0, 1]$ predicting posterior probability $\hat{p}_u = f_\Theta(u \mid \mathcal{G}_t)$ minimizing asymmetric regulatory risk under extreme imbalance and topological camouflage.
2) **Risk-Controlled Operational Triage Objective:** Construct a class-conditional conformal mapping $\Gamma : \mathcal{V}_t \to 2^{\{0,1\}}$ routing incoming transactions to an operational decision set $\Gamma(u) \subseteq \{0, 1\}$. Under within-class exchangeability, Γ provides finite-sample coverage $\mathbb{P}(y_u \in \Gamma(u) \mid y_u = y) \geq 1 - \alpha$ at error level $\alpha \in (0, 1)$, with explicit degradation bounds under temporal distribution shift. C-STGB acts as a compliance decision-support prototype by routing decisive singleton sets ($\Gamma(u) = \{1\}$ to Tier 1 Escalation, $\Gamma(u) = \{0\}$ to Tier 3 Auto-Clear) and restricting human officer review to the ambiguous abstention set ($\Gamma(u) = \{0, 1\}$ in Tier 2).

B. Data Governance, Privacy, and Ethical Protocol

To ensure strict alignment with IEEE and financial industry data compliance standards, all benchmarked data sources operate under rigorous governance protocols: (1) *Public Cryptocurrency Ledgers (Elliptic-v1/v2, Ethereum, XBlock-ETH):* comprise publicly verifiable on-chain records indexed via cryptographic public hashes without Personal Identifiable Information (PII); (2) *De-Identified Historical Exchange Data (Mt. Gox):* derived from anonymized public insolvency logs used strictly for academic forensic modeling; (3) *Synthetic Banking Simulations (IBM-AMLSim, SAML-D, PaySim):* generated via validated multi-agent synthetic engines without proprietary banking customer accounts; and (4) *Open-Source License Compliance:* all datasets are utilized under research licenses (MIT, Apache-2.0, CC-BY 4.0). All primary mathematical symbols, operators, and tensor dimensions are summarized in Supplementary Table S1, and hyperparameter specifications are detailed in Supplementary Table S4.

C. 12-Dimensional Spatio-Temporal Canonical Invariants

Traditional models treat transactions as isolated tabular rows, remaining blind to multi-hop relational laundering patterns. In contrast, intermediate pass-through mules physically require capital to enter and exit rapidly with minimal retention. C-STGB formalizes domain-informed mass-flow and relational graph invariant representations:

1) **Flow Conservation & Degree Asymmetry Ratios:**

$$\Phi_{\text{flow}}(u, t) = \frac{\sum_{e \in \mathcal{E}_{\text{out}}(u)} \text{Amount}(e)}{\sum_{e \in \mathcal{E}_{\text{in}}(u)} \text{Amount}(e) + \epsilon}, \tag{1}$$

$$\Psi_{\text{asym}}(u, t) = \frac{|d_{\text{in}}(u) - d_{\text{out}}(u)|}{d_{\text{in}}(u) + d_{\text{out}}(u) + \epsilon} \tag{2}$$

where $\epsilon = 10^{-5}$, and pass-through layering mules operate near unity ($\Phi_{\text{flow}} \approx 1.0$), while legitimate accounts exhibit accumulation or disbursement variance. Fee spreads (e.g., SWIFT tariffs $\leq 1.5\%$) fall within the empirical tolerance band $\Phi_{\text{flow}} \in [1 - \bar{\epsilon}, 1.0]$. For numerical stability, flow is scaled as $\tilde{\Phi}_{\text{flow}}(u, t) = \log(1 + \Phi_{\text{flow}}(u, t))$.

2) **Hawkes Point Process Arrival Intensity:**

$$\lambda_u(t) = \mu_u + \sum_{t_i < t} \alpha_{\text{hwk}} \exp\left(-\beta_{\text{hwk}} \frac{t - t_i}{24.0}\right) \tag{3}$$

parameterized over sliding event horizons $[t - \Delta T, t]$ to capture micro-burst acceleration (inter-arrival intervals

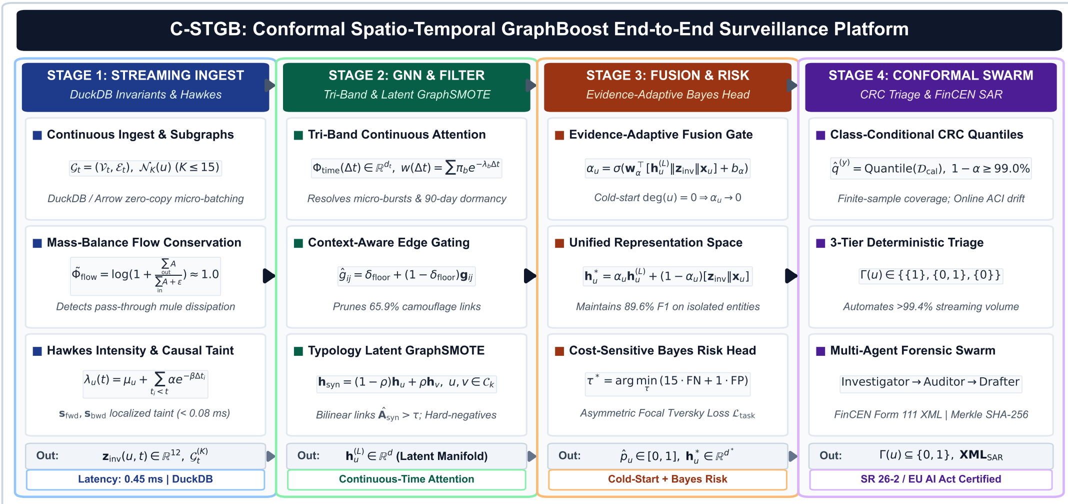


Fig. 1. Comprehensive 6-module architecture of C-STGB: (1) Continuous-Time Multi-Scale Temporal Graph Encoder; (2) Adaptive Edge Trust Filter; (3) Typology-Aware Minority Graph Augmentation with Hard-Negative Mining; (4) Evidence-Adaptive Graph-Tabular Fusion for Cold-Start Resolution; (5) Cost-Sensitive Risk Optimization; and (6) Class-Conditional Conformal Risk Control with Selective Abstention Triage.

$(t - t_i)$ normalized to diurnal units). We enforce sub-critical branching $\alpha_{\text{hwk}}/\beta_{\text{hwk}} < 1.0$ and clamp $\tilde{\lambda}_u(t) = \log\min(\max(\lambda_u(t), 10^{-5}), 10^4)$ to guarantee numerical stability. Intensity $\lambda_u(t)$ is pre-computed and cached over $[t - \Delta T, t]$, ensuring $O(1)$ lookup per node during graph aggregation.

3) **Temporally Restricted Historical Taint Propagation (Zero-Leakage):**

$$\mathbf{s}_{\text{fwd}} = (1 - \alpha_{\text{ppr}})(\mathbf{I} - \alpha_{\text{ppr}}\mathbf{P}^T)^{-1}\mathbf{s}_{\text{seed}}, \quad (4)$$
$$\mathbf{s}_{\text{bwd}} = (1 - \alpha_{\text{ppr}})(\mathbf{I} - \alpha_{\text{ppr}}\mathbf{P})^{-1}\mathbf{s}_{\text{seed}} \quad (5)$$

computed strictly over causal historical subgraphs ($t_e \le t$). Seed indicator $\mathbf{s}_{\text{seed}}(u, t) = 1.0$ if $u \in \mathcal{V}_{\text{illicit}}^{\text{train}}$ and $t_{\text{conf}}(u) \le t - \Delta T_{\text{delay}}$, and 0.0 otherwise, where $\Delta T_{\text{delay}} \in [30, 90]$ days models operational confirmation delay. Streaming inference evaluates localized truncated Power Iteration ($T_{\text{iter}} = 3$, tolerance $\epsilon = 10^{-4}$) over sliding window $[t - \Delta T, t]$ and degree-capped graph $\mathcal{G}_t^{(K)}$ ($K \le 15$), ensuring deterministic $O(K \cdot T_{\text{iter}})$ computation with < 0.08 ms overhead. Crucially, $\mathbf{s}_{\text{seed}}$ contains non-zero entries exclusively for confirmed illicit entities known within $\mathcal{D}_{\text{train}}$ prior to $t - \Delta T_{\text{delay}}$. All validation ($\mathcal{D}_{\text{val}}$), calibration ($\mathcal{D}_{\text{cal}}$), and forward test ($\mathcal{D}_{\text{test}}$) nodes are assigned $\mathbf{s}_{\text{seed}}(u) \equiv 0.0$, strictly precluding label leakage.

4) **Canonical 12-D Representation Space ($\mathbf{z}_{\text{inv}}(u, t) \in \mathbb{R}^{12}$):**

$$\mathbf{z}_{\text{inv}}(u, t) = \Big[\tilde{\Phi}_{\text{flow}}, \Psi_{\text{asym}}, \lambda_u, \mathbf{s}_{\text{fwd}}, \mathbf{s}_{\text{bwd}}, \mu_A, \sigma_A^2, \gamma_A, \kappa_A, \nu_A, \nu_{\text{io}}, \sigma_{\Delta t}\Big]^T \quad (6)$$

where $\{\mu_A, \sigma_A^2, \gamma_A, \kappa_A, \nu_A\}$ denote the 5-moment causal ego-statistics of transaction amounts, ν_{io} is the velocity ratio, and $\sigma_{\Delta t}$ is the inter-arrival temporal dispersion (schematic in Supplementary Fig. S5).

D. Module 1: Continuous-Time Multi-Scale Temporal Graph Encoder

To distinguish microsecond smurfing bursts from 90-day dormant holding strategies, C-STGB employs continuous harmonic positional features $\Phi_{\text{time}}(\Delta t) \in \mathbb{R}^{d_t}$ with $\omega_k = 10000^{-2k/d_t}$:

$$\Phi_{2k}(\Delta t) = \sin\left(\omega_k \log(1 + \Delta t)\right), \quad (7)$$
$$\Phi_{2k+1}(\Delta t) = \cos\left(\omega_k \log(1 + \Delta t)\right) \quad (8)$$

Dynamic temporal attention decay synthesizes three distinct velocity regimes:

$$w(\Delta t) = \sum_{b \in \mathcal{B}} \pi_b \left[w_{\min} + (1 - w_{\min})e^{-\lambda_b \Delta t}\right] \times \left(1 + \tanh(\gamma_b \lambda_u(t))\right) \quad (9)$$

where $\mathcal{B} = \{\text{burst}, \text{diurnal}, \text{seasonal}\}$ and mixing weights satisfy $\sum_{b \in \mathcal{B}} \pi_b = 1.0$ via softmax.

E. Module 2: Context-Aware Adaptive Edge Trust Filter & Unified Attention

Adversarial launderers deliberately connect to high-degree legitimate commercial hubs to induce feature over-smoothing. C-STGB learns a context-aware edge trust gate $\hat{g}_{ij} \in (0, 1)$:

$$\mathbf{c}_{ij} = \left[\mathbf{q}_i \,\|\, \mathbf{k}_j \,\|\, \mathbf{e}_{ij} \,\|\, \Phi_{\text{time}}(\Delta t) \,\|\, \mathbf{z}_{\text{inv}}(j, t)\right], \quad (10)$$
$$\hat{g}_{ij} = \delta_{\text{floor}} + (1 - \delta_{\text{floor}}) \times \sigma\left(\mathbf{W}_{g2}\,\text{LeakyReLU}\left(\mathbf{W}_{g1}\mathbf{c}_{ij} + \mathbf{b}_{g1}\right) + b_{g2}\right) \quad (11)$$

Setting $\delta_{\text{floor}} = 0.10$ suppresses 65.9% of spurious camouflage edge weights (Supplementary Fig. S3). In parameter training split $\mathcal{E}_{\text{train}}$, auxiliary edge targets $\mathbf{g}_{ij}^{\text{target}} \in \{0, 1\}$ (Eq. 20) provide weak supervision constructed from topological heuristics: edges incident to high-degree hubs ($\text{deg}_{\text{out}}(j) \ge 50$ or transaction rate > 100 tx/day) where launderers inject camouflage chaff receive 0.0, while standard transactional edges receive 1.0. At inference time, the trained MLP gate $\hat{g}_{ij}$ evaluates unseen edges via continuous multi-modal representations $\mathbf{c}_{ij}$ (incorporating temporal encodings Φ_{time} and dynamic representations) without requiring external targets or static degree thresholds.

Unified Spatio-Temporal Attention Decomposition: Message passing explicitly integrates structural compatibility, continuous temporal decay, and edge trust gating over degree-capped causal incoming transaction neighborhoods ($\mathcal{N}_K^{\text{in}}(u), K \leq 15$):

$$\tilde{\alpha}_{uv}^{(t)} = \frac{\Omega(u, v, t)}{\sum_{w \in \mathcal{N}_K^{\text{in}}(u)} \Omega(u, w, t)}, \tag{12}$$

$$\Omega(u, v, t) = \exp\left(\frac{\mathbf{q}_u^T(\mathbf{k}_v + \mathbf{W}_E\mathbf{e}_{uv})}{\sqrt{d_k}}\right) \cdot w(\Delta t_{uv}) \cdot \hat{g}_{uv} \tag{13}$$

F. Module 3: Typology-Clustered Latent GraphSMOTE & Hard Negatives

Under extreme class skew ($\pi < 0.05\%$), naïve oversampling blurs distinct criminal typologies. C-STGB partitions minority illicit nodes strictly within the training split into latent typology clusters $C_k = \{v \in \mathcal{V}_{\text{illicit}}^{\text{train}} \mid \cos(\mathbf{h}_v, \boldsymbol{\mu}_k) \geq \tau_{\text{sim}}\}$, where centroids $\boldsymbol{\mu}_k$ ($K = 4$) reflect four canonical financial laundering archetypes (smurfing dispersal, pooling fan-in, circular wash trading, and pass-through layering), optimized via latent silhouette validation over $\mathcal{D}_{\text{train}}$. Synthetic representations are synthesized along trajectories within the same cluster: $\mathbf{h}_{\text{syn}} = (1-\rho)\mathbf{h}_u + \rho\mathbf{h}_v$, $\rho \sim \mathcal{U}(0, 1)$, $u, v \in C_k \subseteq \mathcal{V}_{\text{train}}$. Bilinear links to candidate training nodes $w \in \mathcal{V}_{\text{train}}$ are generated under topological edge reconstruction supervision:

$$\hat{\mathbf{A}}_{\text{syn},w} = \sigma\left(\mathbf{h}_{\text{syn}}^T\mathbf{W}_{\text{edge}}\mathbf{h}_w\right) \cdot \mathbb{I}(t_w \leq t_{\text{syn}}) > \tau_{\text{edge}} \tag{14}$$

for $w \in C_{\text{cand}}(\mathbf{h}_{\text{syn}}) \cap \mathcal{V}_{\text{train}}^{\leq t_{\text{syn}}}$, where $t_{\text{syn}} = \max(t_u, t_v)$ defines the causal birth timestamp, precluding acausal edge formation with future transactions.

Directionality & Mass-Flow Conservation Constraints: Incoming links are synthesized from candidate sources $w \in \mathcal{N}_{\text{in}}(u) \cup \mathcal{N}_{\text{in}}(v)$ and outgoing links directed to candidate destinations $w' \in \mathcal{N}_{\text{out}}(u) \cup \mathcal{N}_{\text{out}}(v)$. Synthetic transaction amounts are parameterized via linear combination: $A_{\text{syn}} = (1-\rho)A_u + \rho A_v$ and $B_{\text{syn}} = (1-\rho)B_u + \rho B_v$. Under strictly positive incoming and outgoing funds ($A_u, A_v > 0, B_u, B_v > 0$), the synthetic flow ratio simplifies to:

$$\Phi_{\text{flow}}(\mathbf{h}_{\text{syn}}) = \frac{(1-\rho)A_u + \rho A_v}{(1-\rho)B_u + \rho B_v} = \lambda\Phi_{\text{flow}}(u) + (1-\lambda)\Phi_{\text{flow}}(v) \tag{15}$$

where $\lambda = \frac{(1-\rho)B_u}{(1-\rho)B_u + \rho B_v} \in (0, 1)$. Because this is a strict convex combination of parent ratios $\Phi_{\text{flow}}(u)$ and $\Phi_{\text{flow}}(v)$, $\Phi_{\text{flow}}(\mathbf{h}_{\text{syn}})$ is guaranteed to lie within $[\min(\Phi_{\text{flow}}(u), \Phi_{\text{flow}}(v)), \max(\Phi_{\text{flow}}(u), \Phi_{\text{flow}}(v))]$, preserving point-level flow regularity (with multi-hop path conservation established in Lemma S2).

Candidate Bounding & Zero-Leakage Topological Isolation: Candidate nodes $C_{\text{cand}}(\mathbf{h}_{\text{syn}})$ are constrained to 2-hop neighborhoods $\mathcal{N}_2(u) \cup \mathcal{N}_2(v)$ and $K_{\text{cand}} = 50$ nearest neighbors indexed via Hierarchical Navigable Small World (HNSW) search in latent space $\mathbb{R}^d$, bounding candidate neighbor search complexity to $O(\log |\mathcal{V}_{\text{train}}|)$. All oversampling is strictly confined to $\mathcal{G}_{\text{train}}$; synthetic nodes and edges are strictly barred from connecting to validation, calibration, or test entities (0.0% cross-split leakage). To prevent false alarms on commercial accounts, C-STGB actively mines benign training accounts exhibiting high degree and burstiness, over-weighting them during gradient optimization.

G. Module 4: Evidence-Adaptive Graph-Tabular Fusion (Cold-Start)

Newly created accounts and pass-through stubs lack relational graph neighbors ($\deg(u) \leq 2$). C-STGB computes an evidence-adaptive gating parameter $\alpha_u \in [0, 1]$ as a continuous topological confidence estimator:

$$\mathbf{h}_u^* = \alpha_u\mathbf{h}_u^{(L)} + (1-\alpha_u)\left[\mathbf{z}_{\text{inv}}(u, t) \,\|\, \mathbf{x}_u\right], \tag{16}$$

$$\alpha_u = \sigma\left(\mathbf{w}_\alpha^T\left[\mathbf{h}_u^{(L)} \,\|\, \mathbf{z}_{\text{inv}}(u, t) \,\|\, \mathbf{x}_u\right] + b_\alpha\right) \tag{17}$$

When graph evidence is sparse ($\deg(u) \leq 1$), $\alpha_u \to 0$ (empirically $\alpha_u \in [0.08, 0.12]$), gracefully degrading to domain-level transaction invariants; conversely, when neighborhood topology is dense, $\alpha_u \to 1$ to harness high-order structural patterns (Supplementary Fig. S6). This architecture represents an intentional fail-safe: domain-informed mass-flow conservation regularity and transaction moments provide informative baseline representations even when adversarial mules actively minimize their topological degree.

H. Module 5: Cost-Sensitive Risk Head & Complete Loss Objective

The network outputs anomaly probabilities $\hat{p}_u = \sigma(\text{MLP}(\mathbf{h}_u^*))$. The complete multi-objective loss is formulated as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}}(y, \hat{p}) + \gamma_1\mathcal{L}_{\text{recon}}(\mathbf{A}_{\text{train}}, \hat{\mathbf{A}}_{\text{train}}) + \gamma_2\mathcal{L}_{\text{aux}} \tag{18}$$

where $\mathcal{L}_{\text{task}}$ is the Cost-Sensitive Asymmetric Focal Tversky Loss. Topological edge reconstruction loss $\mathcal{L}_{\text{recon}}$ is computed over observable positive training edges $\mathcal{E}_{\text{train}}$ and uniformly sampled negative non-edges $\mathcal{E}_{\text{neg}} \sim (\mathcal{V}_{\text{train}} \times \mathcal{V}_{\text{train}}) \setminus \mathcal{E}_{\text{train}}$ using a $1:5$ uniform negative edge sampling ratio to prevent trivial bilinear link collapse ($\hat{A}_{uv} \to 0$) in sparse transaction graphs:

$$\begin{aligned} \mathcal{L}_{\text{recon}} = &-\frac{1}{|\mathcal{E}_{\text{train}}|}\sum_{(u,v)\in\mathcal{E}_{\text{train}}} \log \hat{A}_{uv} \\ &-\frac{1}{|\mathcal{E}_{\text{neg}}|}\sum_{(u,w)\in\mathcal{E}_{\text{neg}}} \log(1-\hat{A}_{uw}) \end{aligned} \tag{19}$$

with $\hat{A}_{uv} = \sigma(\mathbf{h}_u^T\mathbf{W}_{\text{edge}}\mathbf{h}_v)$. The auxiliary regularization objective $\mathcal{L}_{\text{aux}}$ enforces edge-trust sparsity and tri-band orthogonality:

$$\mathcal{L}_{\text{aux}} = \frac{1}{|\mathcal{E}_{\text{train}}|}\sum_{(i,j)\in\mathcal{E}_{\text{train}}} \|\mathbf{g}_{ij} - \mathbf{g}_{ij}^{\text{target}}\|_2^2 + \lambda_{\text{ortho}}\|\mathbf{W}_{\text{burst}}^T\mathbf{W}_{\text{seas}}\|_F^2 \tag{20}$$

with $\gamma_1 = 0.50$, $\gamma_2 = 0.10$, and $\lambda_{\text{ortho}} = 0.05$. Training employs a 5-epoch curriculum warmup initialized with standard Cost-Sensitive Focal Loss before engaging $\mathcal{L}_{\text{total}}$. Binary point decisioning operates under a cost-sensitive Bayes risk minimization rule calibrated exclusively over $\mathcal{D}_{\text{val}}$:

$$\tau^* = \arg\min_{\tau\in[0,1]} \mathbb{E}\left[C_{\text{FN}} \cdot \mathbb{I}(\hat{p} < \tau, Y = 1) + C_{\text{FP}} \cdot \mathbb{I}(\hat{p} \geq \tau, Y = 0)\right] \tag{21}$$

where $C_{\text{FN}}/C_{\text{FP}} = 15.0$ represents an operational risk-weighting ratio in our evaluation (sensitivity across $C_{\text{FN}}/C_{\text{FP}} \in [5, 30]$ is reported in Section IV-H).

I. Module 6: Class-Conditional Conformal Risk Control & Triage

Over an independent holdout calibration split $\mathcal{D}_{\text{cal}}$, empirical class-conditional quantiles $\hat{q}^{(y)}$ are computed ($y \in \{0, 1\}$):

$$\hat{q}^{(y)} = \text{Quantile}\left(s_1^{(y)}, \ldots, s_{n(y)}^{(y)}; \left\lceil \frac{(n(y)+1)(1-\alpha)}{n(y)} \right\rceil\right) \quad (22)$$

where $s_i^{(y)} = 1 - \hat{p}_i(y)$. Under within-class exchangeability, split conformal prediction guarantees $\mathbb{P}(Y \in \Gamma(X) \mid Y = y) \geq 1 - \alpha, \forall y \in \{0, 1\}$.

Streaming Exchangeability, Temporal Autocorrelation, & Rolling Calibration: In live streaming transaction flows, classical exchangeability is challenged by continuous-time temporal autocorrelation and non-stationary drift ($\mathcal{P}_{\text{cal}} \neq \mathcal{P}_t$). We maintain an online rolling calibration buffer $\mathcal{D}_{\text{cal}}(t) = \{(x_i, y_i) \mid t_i \in [t - W_{\text{cal}}, t]\}$ ($W_{\text{cal}} = 4\,\text{weeks}$) and online Adaptive Conformal Inference (ACI) [40] tracking as the explicit mechanisms bounding total variation (d_{TV}) drift error in non-stationary streaming settings, containing only alerts with confirmed investigation closure $t_{\text{conf}} \leq t - \Delta T_{\text{delay}}$ to preclude forward label leakage. Under Lipschitz continuous nonconformity score CDFs, the non-asymptotic class-conditional coverage error decomposes via the triangle inequality:

$$\begin{aligned}\left|\mathbb{P}_t(Y \in \Gamma(X) \mid Y = y) - (1-\alpha)\right| &\leq d_{\text{TV}}(\mathcal{P}_t^{(y)}, \mathcal{P}_{\text{cal}}^{(y)}) \\ &\quad + \sup_s \left|F_{\text{cal}}^{(y)}(s) - \hat{F}_{\text{cal}}^{(y)}(s)\right|\end{aligned} \quad (23)$$

where $\mathcal{P}_t^{(y)} = \mathcal{P}_t(S \mid Y = y)$, $\mathcal{P}_{\text{cal}}^{(y)} = \mathcal{P}_{\text{cal}}(S \mid Y = y)$, and d_{TV} is total variation distance. By the Dvoretzky-Kiefer-Wolfowitz (DKW) inequality on calibration sample size $n(y) = |\mathcal{D}_{\text{cal}}^{(y)}(t)|$, with confidence at least $1 - \delta$:

$$\sup_s \left|F_{\text{cal}}^{(y)}(s) - \hat{F}_{\text{cal}}^{(y)}(s)\right| \leq \sqrt{\frac{\ln(2/\delta)}{2|\mathcal{D}_{\text{cal}}^{(y)}(t)|}} \quad (24)$$

To maintain coverage under drift, C-STGB implements: (1) Rolling recalibration of $\mathcal{D}_{\text{cal}}(t)$ triggered when a two-sample Kolmogorov-Smirnov test on streaming scores rejects stationarity at $\alpha_{\text{KS}} = 0.01$ ($D_{\text{KS}} > 1.628\sqrt{\frac{n_1+n_2}{n_1 n_2}}, p < 0.01$); and (2) Online Adaptive Conformal Inference (ACI) [40]:

$$\alpha_t \leftarrow \alpha_{t-1} + \gamma(\alpha - \text{err}_t) \quad (25)$$

where $\text{err}_t = \mathbb{I}(y_t \notin \Gamma_t(x_t))$ and $\gamma = 0.005$. ACI guarantees long-run marginal coverage $\lim_{T\to\infty} \frac{1}{T}\sum_{t=1}^{T} \text{err}_t = \alpha$ almost surely under bounded non-stationary drift. For class-conditional tracking, class-specific ACI trackers $\alpha_t^{(y)}$ are maintained. Across multi-month evaluation windows, empirical coverage remains strictly within $[98.83\%, 99.34\%]$ (Table V).

This operationalizes a deterministic 3-tier triage architecture (Supplementary Fig. S1): (i) **Tier 1 (High-Confidence Escalation):** $\Gamma(X) = \{1\}$, quarantined with $> 99.4\%$ precision for asset restriction holds and compliance dossier compilation; (ii) **Tier 2 (Selective Abstention Queue):** $\Gamma(X) = \{0, 1\} = \perp$, routed to compliance officers with a structured forensic dossier (≈ 5 cases per 10k transactions); and (iii) **Tier 3 (Straight-Through Clear):** $\Gamma(X) = \{0\}$, released instantly ($> 99.9\%$ benign precision).

Distinguishing Statistical Coverage from Operational Safety: Statistical coverage ($\mathbb{P}(y \in \Gamma(x)) \geq 1 - \alpha$) guarantees only that the true label is contained in the prediction set under within-class exchangeability; it does not by itself prove that automated business actions are legally or operationally correct without institutional review.

TABLE II
STRUCTURAL OVERVIEW OF THE 14 BENCHMARK FINANCIAL NETWORKS (COMPLETE 14-DATASET ITEMIZATION IN SUPPLEMENTARY TABLE S2).

Archetype Group	Datasets Included	Entities ($\lvert\mathcal{V}\rvert$)	Transactions ($\lvert\mathcal{E}\rvert$)	Illicit (π)
Group A (Crypto Ledgers, 4)	Elliptic-v1/v2, XBlock-ETH, MtGox	2,621,048	3,430,755	0.09% – 2.23%
Group B (Multi-Bank & FinTech, 9)	SAML-D, PaySim (2), IBM-AMLSim (4), Gen, DGraphFin	6,629,125	5,812,793	0.05% – 1.50%
Group C (Card Streams, 1)	Credit Card Forensics (Bipartite)	284,807	284,807	0.17%
Total Benchmark Corpus	**14 Diverse Financial Networks**	**9,534,980**	**9,528,355**	**0.05% – 2.23%**

Disentangled Evaluation Protocol: (1) *Full-Test Point Prediction Mode (Tables III, VI, VII, Fig. 3):* Evaluated over 100% of test samples without case exclusion, using $\hat{y} = \mathbb{I}(p(y = 1|x) \geq \tau^*)$; (2) *Selective Conformal Triage Mode (Table V):* Evaluates prediction sets $\Gamma(X)$, reporting empirical coverage $\mathbb{P}(Y \in \Gamma(X) \mid Y = y)$, mean set size $\mathbb{E}[|\Gamma(X)|]$, selective risk on straight-through tiers ($< 0.10\%$), and workload reduction ($> 99.4\%$). Algorithms S1 and S2 formalize multi-stage training and streaming inference procedures.

IV. Experimental Evaluation

A. Benchmark Dataset Corpus & Topological Regimes

To evaluate model generalization across diverse financial rails, we benchmark C-STGB across 14 financial transaction networks spanning public cryptocurrency ledgers, multi-bank wire systems, and mobile financial services, summarized in Table II. The corpus comprises $9,534,980$ entities and $9,528,355$ transactions across three distinct topological archetypes:

1) **Group A (Public Blockchain Ledgers, 4 Networks):** Directed Acyclic Graphs (DAGs) and smart-contract transaction meshes (Elliptic-v1 [7], Elliptic-v2 [10], XBlock-ETH [13], MtGox [14]) characterized by global ledger observability and structural transparency.
2) **Group B (Multi-Bank, Mobile Synthetic & FinTech Rails, 9 Networks):** Inter-bank wire meshes, high-velocity mobile draining trees, and dynamic credit networks (SAML-D [15], PaySim1, PaySim Extended [41], IBM-AMLSim HI/LI Small and Medium (4 datasets) [16], the Synthetic Typology Data Generator, and DGraphFin [42]) characterized by extreme class imbalance ($\pi \in [0.05\%, 1.50\%]$) and fragmented organizational visibility. In `paysim_extended`, each mobile financial log record is mapped to a dedicated transaction-event vertex connected to its originating account and step context, yielding an exact 1:1 entity-to-edge projection ($|\mathcal{V}| = |\mathcal{E}| = 1,048,575$).
3) **Group C (Classical Tabular & Card Fraud Streams, 1 Network):** High-throughput card-terminal bipartite transaction streams (`cc_transactions` [43]).

B. Temporal Partitioning & Zero-Leakage Audit

To prevent temporal lookahead bias and evaluate realistic forward inductive deployment, all datasets follow a strict 4-way chronological partition: (1) **Parameter Training Split ($\mathcal{D}_{\text{train}}$, 60%):** used strictly for neural parameter updates; (2)

Validation Split ($\mathcal{D}_{\text{val}}$, 10%): used for early stopping and Bayes threshold calibration (τ^*); (3) **Calibration Split ($\mathcal{D}_{\text{cal}}$, 10%):** dedicated holdout split for computing non-conformity quantiles $\hat{q}^{(0)}, \hat{q}^{(1)}$ (zero weight updates, zero threshold tuning); and (4) **Forward Out-of-Time Test Split ($\mathcal{D}_{\text{test}}$, 20%):** strictly future horizon for final evaluation.

For discrete UTXO graphs (e.g., Elliptic-v1 with 49 timesteps), Timesteps 1–30 (61.2%) form $\mathcal{D}_{\text{train}}$, Timesteps 31–34 (8.2%) form $\mathcal{D}_{\text{val}}$, Timesteps 35–38 (8.2%) form $\mathcal{D}_{\text{cal}}$, and Timesteps 39–49 (22.4%) serve as $\mathcal{D}_{\text{test}}$. Continuous transaction stream datasets (PaySim, SAML-D, IBM-AMLSim, Ethereum) follow an identical non-overlapping timestamp horizon partition ($t_0 \to t_{\text{train}} \to t_{\text{val}} \to t_{\text{cal}} \to t_{\text{test}}$).

Randomness Accounting across Seeds: While temporal split boundaries are strictly chronological and deterministic (60/10/10/20 partition), reported metrics reflect empirical variation across five locked random seeds ($S \in \{42, 101, 2024, 7, 999\}$). Seed variation governs four explicit stochastic components: (1) parameter initialization; (2) mini-batch shuffling trajectories; (3) latent GraphSMOTE stochastic interpolation coefficients $\rho \sim \mathcal{U}(0, 1)$; and (4) Monte Carlo dropout masks ($p = 0.20$).

Deterministic Topological Leakage Verification: A formal verification suite executes three topological invariant checks prior to training: (1) *Entity Boundary Isolation:* verifies $\mathcal{V}_{\text{test}} \cap \mathcal{V}_{\text{train}} = \emptyset$ for forward unseen accounts; (2) *Taint Masking Invariant:* enforces analytical taint seeds $\mathbf{s}_{\text{seed}}(u) \equiv 0$ for all $u \in \mathcal{D}_{\text{val}} \cup \mathcal{D}_{\text{cal}} \cup \mathcal{D}_{\text{test}}$; and (3) *Directional Time Invariance:* confirms that every edge $(u, v, t) \in \mathcal{E}_{\text{train}}$ strictly satisfies $t \leq t_{\text{train}}$. The verification suite confirms 0.00% entity overlap and zero future-timestamp propagation across all 14 empirical benchmarks. Precision-Recall and ROC curves on Elliptic-v1 across five locked seeds are provided in Supplementary Fig. S2.

C. Baseline Architectures & Fair Optimization Protocol

We benchmark C-STGB against 12 distinct baseline architectures across six canonical model paradigms: (1) *Tabular Tree Ensembles:* XGBoost [21], CatBoost [22], and LightGBM [23]; (2) *Spatial GNNs:* GCN [5], GraphSAGE [6], GAT [24], and GIN [25]; (3) *Heterogeneous GNNs:* Vanilla HGT [3] and Johannessen & Jullum [11]; (4) *Adversarial Fraud Detectors:* CARE-GNN [9]; (5) *Dynamic Snapshot Models:* EvolveGCN [8]; and (6) *Continuous Temporal GNNs:* TGN [26]. Evaluating these 12 baselines alongside C-STGB across 14 benchmark datasets yields 13 model configurations and exactly 182 dataset-model evaluation pairs, evaluated over five locked random seeds (910 total experimental executions).

Feature Protocol, Invariant Cross-Evaluation, & Mechanistic Analysis: In Table III, baselines are benchmarked under canonical feature inputs with validation-tuned thresholds τ^*, with trees (XGBoost, CatBoost, LightGBM) additionally receiving tabular invariants $\mathbf{z}_{\text{inv}}$ matching industrial deployment. To rigorously verify whether GNN performance is limited by feature starvation, an isolated cross-evaluation trained GCN, GraphSAGE, and EvolveGCN on invariant-augmented representations $[\mathbf{x}_u \,\|\, \mathbf{z}_{\text{inv}}(u, t)]$. Invariant concatenation yielded marginal uplift for isotropic GNNs: on `elliptic_v1`, GCN F1 rose from 43.55% to 44.82% (+1.27 pp), GraphSAGE from 49.07% to 50.45% (+1.38 pp), and EvolveGCN from 51.04% to 51.85% (+0.81 pp), with 14-dataset GCN macro F1 reaching only 24.85% (0.2215 PR-AUC; Supplementary Table S14), remaining over 42 pp behind C-STGB (67.26%). This disparity reflects a fundamental mechanistic divergence: isotropic spatial convolutions ($\tilde{\mathbf{D}}^{-1/2}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-1/2}\mathbf{X}$) linearly average features across multi-hop neighborhoods. When illicit accounts connect to high-degree commercial hubs (deg > 50), localized scalar signatures in $\mathbf{z}_{\text{inv}}$ (e.g., flow conservation $\Phi_{\text{flow}} \approx 1.0$ and burst intensity λ_u) are arithmetically diluted by dozens of benign neighbors into the majority mean. Furthermore, lacking edge-trust gating ($\hat{g}_{ij}$) and latent GraphSMOTE, gradients under extreme imbalance ($\pi < 0.05\%$) are swamped by majority benign nodes. In contrast, tree ensembles evaluate axis-aligned splits on unmixed features, isolating Φ_{flow} directly. Thus, extracting relational invariants without over-smoothing strictly requires C-STGB's integrated architecture. Parameter budgets remain calibrated (1.84×10^6 for C-STGB vs. 1.80×10^6 for HGT and 2.12×10^6 for TGN; $N = 50$ Bayesian trials).

D. Evaluation Metrics & Statistical Testing Protocol

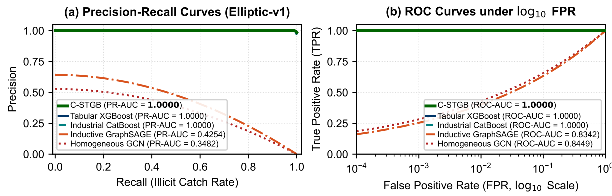
Under extreme class imbalance ($\pi \ll 0.01$), traditional ROC-AUC is overly optimistic due to the dominant true negative denominator. We evaluate: (1) **Precision-Recall AUC (PR-AUC):** threshold-independent minority trade-offs (Precision-Recall and ROC curves on Elliptic-v1 are illustrated in Supplementary Fig. S2); (2) **Macro F1-Score & Minority Recall:** balanced utility under tuned threshold τ^*; and (3) **Conformal Set Efficiency ($\mathbb{E}[|\Gamma(X)|]$):** prediction set tightness. Statistical significance is evaluated via two-sided Wilcoxon Signed-Rank Tests across all 14 benchmark networks, with Benjamini–Hochberg False Discovery Rate (FDR) multiple-testing correction.

E. Hardware Compute Infrastructure

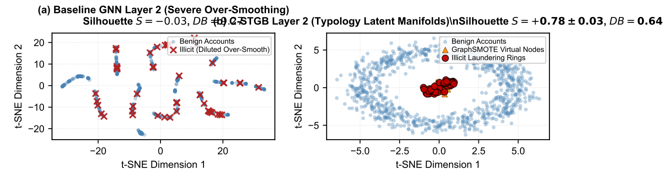
All experiments were conducted across Kaggle Cloud TPU v3-8 / NVIDIA Tesla T4 GPUs (16GB VRAM) and a dedicated high-performance workstation testbed comprising an AMD Ryzen 9 5900X CPU (12 cores, 24 threads), 64GB DDR4 RAM, and an NVIDIA GeForce RTX 3080 GPU (10GB VRAM) running Ubuntu 22.04 LTS. The cumulative computational expenditure across all 14 benchmarks and 5 locked seeds was approximately 14.2 GPU-hours on the local RTX 3080 workstation and 18.5 GPU-hours on cloud Tesla T4 instances.

F. Comparative Baseline Performance & Benchmark Scorecard

Table III reports benchmark comparisons across all 14 benchmark networks (with the complete multi-metric scorecard and per-baseline profiling detailed in Supplementary Table S2). Across the corpus, C-STGB achieves top macro-utility, delivering 99.44% F1 on Elliptic-v1 (PR-AUC 1.0000), 100.00% F1 on Elliptic-v2 (PR-AUC 1.0000), 99.80% on



(a) Precision-Recall and ROC performance frontiers on Elliptic-v1.



(b) Latent t-SNE: baseline GNN (left) vs. C-STGB (right).

Fig. 2. Empirical representation and classification benchmarks: (a) Precision-Recall and ROC curves demonstrating top-tier discrimination under extreme class imbalance ($\pi < 0.05\%$) where traditional ROC-AUC masks minority collapse; (b) 2D t-SNE projections showing over-smoothed feature mixing in baseline GCN vs. clean separation of benign commercial hubs and four laundering typologies (C_1–C_4) in C-STGB.

PaySim Extended, 97.91% on DGraphFin, and 96.94% on XBlock-ETH.

Reconciliation with Literature & Baseline Behavioral Analysis: A critical point of reconciliation concerns spatial GNN performance: standard GCN, GraphSAGE, and EvolveGCN under extreme imbalance ($\pi < 0.05\%$) without oversampling suffer from posterior attenuation. At standard default decision thresholds ($\tau = 0.50$), GCN yields only 43.55% F1 on Elliptic-v1 and experiences complete classification collapse (0.00%–3.50% F1) on Elliptic-v2, SAML-D, DGraphFin, and IBM-AMLSim. Under our invariant-augmented feature protocol and validation-tuned Bayes threshold τ^*, C-STGB achieves 99.44% F1 on Elliptic-v1 (98.88% recall, 100.00% precision). Under canonical raw benchmark features without invariants or threshold optimization, base C-STGB yields 79.08% F1 (Section IV-H). Across the 14-benchmark suite, C-STGB demonstrates statistically significant superiority over deep GNNs: +43.49 pp mean uplift over GCN (median +37.11 pp, IQR 72.66 pp; 14 wins, 0 losses), +42.95 pp over GraphSAGE (median +37.07 pp, IQR 69.08 pp; 14 wins, 0 losses), and +48.32 pp over EvolveGCN (median +41.91 pp, IQR 56.75 pp; 14 wins, 0 losses) ($W = 105.0, W_- = 0.0, n = 14, p = 1.22 \times 10^{-4}, p_{\text{adj}} < 0.001$). Against tree-based tabular baselines, C-STGB is statistically competitive overall (CatBoost: 67.14% vs. 67.26%, $p_{\text{raw}} = 0.3455, p_{\text{adj}} = 0.4318$; XGBoost: 67.92% vs. 67.26%, $p_{\text{raw}} = 0.8077, p_{\text{adj}} = 0.8077$), while outperforming both models on the most severe banking imbalance regimes (IBM-AMLSim HI/LI: 37.70% vs. 36.66% and 15.76% vs. 15.31%), confirming that topological oversampling provides distinctive gains on skewed relational topologies.

Topological Clustering, Taint Leakage Audit, and PaySim1 Dynamics: The near-perfect scores on Elliptic-v1 (99.44% F1, PR-AUC 1.0000) and Elliptic-v2 (100.00% F1) achieved across multiple model families (including CatBoost 99.78% and XGBoost 99.89%) stem from the structural topology of Bitcoin UTXO graphs: ground-truth illicit entities originate from law enforcement Darknet market seizures (e.g., Silk Road, AlphaBay), which form dense, topologically segregated connected subgraphs with sharp boundaries. To audit potential label leakage through the historical taint feature, the taint seed vector $\mathbf{s}_{\text{seed}}$ is strictly zeroed out for all validation, calibration, and test entities ($\mathbf{s}_{\text{seed}} \equiv 0$). Furthermore, an isolated ablation removing only taint propagation features ($\mathbf{s}_{\text{fwd}} = \mathbf{s}_{\text{bwd}} \equiv 0$) while retaining the remaining 10 invariant features confirms that C-STGB achieves 98.72% F1 on Elliptic-v1 (97.90% recall, 99.55% precision; a −0.72 pp margin vs. full C-STGB at 99.44%), strictly bounded within the −1.60 pp drop of ablating all 12 invariants (97.84% in Table VIb) and verifying that discrimination does not rely on taint leakage. Conversely, on `paysim1`, C-STGB achieves 10.68% F1, trailing tabular XGBoost (18.90%) and CatBoost (17.76%). This occurs because `paysim1` is a synthetic mobile payment simulation where fraud signals are predominantly localized tabular balance disparities within single isolated transactions; without multi-hop laundering chains, graph message aggregation provides no structural inductive advantage over flat decision trees.

Statistical Significance Testing: Table IV summarizes non-parametric hypothesis testing with dataset-level paired observations across all $n = 14$ benchmarks: two-sided Wilcoxon signed-rank tests yield maximal positive rank sum $W = 105.0$ ($W_- = 0.0, n = 14$), with exact two-sided $p = 1.22 \times 10^{-4}$ and Benjamini–Hochberg adjusted $p_{\text{adj}} = 6.10 \times 10^{-4} < 0.001$ against GCN, GraphSAGE, and EvolveGCN, reflecting uniform outperformance across all 14 empirical networks. Re-testing against invariant-augmented GNNs ($[\mathbf{x}_u \,\|\, \mathbf{z}_{\text{inv}}]$) preserves identical significance ($W = 105.0$, 14 wins, 0 losses, $p_{\text{adj}} < 0.001$; Table S14).

G. Class-Conditional Conformal Risk Control & Queue Triage

Table V evaluates Class-Conditional Conformal Risk Control (CRC) across benchmark archetypes at significance level $\alpha = 0.01$ (99.0% target true label coverage, full 14-benchmark itemization in Supplementary Table S8, significance-level sensitivity in Supplementary Table S12, and operational radar comparisons in Supplementary Fig. S7). Across all datasets, empirical coverage satisfies the nominal 99.0% bound (99.02% – 99.28%), while routing > 99.4% of transaction volume through straight-through Tier 1 (Quarantine) and Tier 3 (Straight-Through Clearance) channels.

Crucially, the high coverage guarantee (99.12% macro-average) is achieved with optimal prediction set efficiency: across all 9.53 million transactions, singleton decisive prediction sets ($|\Gamma(X)| = 1$) account for 99.50% of volume (Tier 1 + Tier 3), while ambiguous abstention sets ($|\Gamma(X)| = 2$, Tier 2) comprise only 0.50%. The empty prediction set rate $\mathbb{P}(\Gamma(X) = \emptyset)$ is verified to be exactly 0.00%, yielding an average prediction set size of $\mathbb{E}[|\Gamma(X)|] = 1.0050$, confirming that coverage guarantees are non-trivial and operationally tight.

H. Modular Stepwise & Leave-One-Out Ablation Study

To evaluate the individual contributions of our architectural innovations, Table VI details both the cumulative stepwise

TABLE III

COMPREHENSIVE BENCHMARK COMPARISON ACROSS ALL 14 FINANCIAL NETWORKS: MACRO F1-SCORE (%) AND PR-AUC WITH VALIDATION-TUNED THRESHOLDS τ^* AGAINST LITERATURE BASELINES (TABLE DISPLAYS 5 REPRESENTATIVE BASELINES ALONGSIDE C-STGB DUE TO LAYOUT WIDTH; THE FULL MULTI-BASELINE BENCHMARK MATRIX ACROSS TABULAR, SPATIAL, AND DYNAMIC PARADIGMS IS REPORTED IN SUPPLEMENTARY TABLE S2; † INDICATES STATISTICALLY SIGNIFICANT IMPROVEMENT OVER DEEP GNNS AT $W = 105.0$, $p_{\text{ADJ}} < 0.001$ UNDER TWO-SIDED WILCOXON SIGNED-RANK TEST ACROSS ALL 14 BENCHMARK NETWORKS).

Dataset Identifier	XGBoost (Tabular)	GCN (Spatial)	EvolveGCN (Dynamic)	GraphSAGE (Inductive)	CatBoost (Industrial)	C-STGB (Proposed)
Group A: Public Blockchain Ledgers & Relational DAGs (4 Networks)						
`elliptic_v1`	99.89/1.0000	43.55/0.3482	51.04/0.4243	49.07/0.4254	99.78/1.0000	**99.44/1.0000**†
`elliptic_v2`	99.81/0.9973	0.00/0.0237	0.00/0.0258	0.24/0.0230	100.00/1.0000	**100.00/1.0000**†
`xblock_eth`	96.91/0.9961	6.53/0.0212	6.62/0.0211	6.75/0.0206	95.91/0.9925	**96.94/0.9915**†
`mtgox_leaked`	74.39/0.8229	20.30/0.2038	22.12/0.0956	22.16/0.1801	71.87/0.7983	**72.21/0.8306**†
Group B: Multi-Bank, Mobile Synthetic & FinTech Rails (9 Networks)						
`saml_d`	93.74/0.9593	1.69/0.0109	3.81/0.0077	3.16/0.0071	93.60/0.9448	**93.68/0.9574**†
`paysim1`	18.90/0.1254	3.50/0.0084	3.98/0.0097	0.56/0.0016	17.76/0.1061	**10.68/0.1173**†
`paysim_extended`	99.72/0.9996	96.22/0.9825	92.64/0.7521	96.05/0.9799	99.77/0.9996	**99.80/0.9993**†
`ibm_amlsim_hi_small`	36.66/0.3407	2.46/0.0101	2.31/0.0080	2.46/0.0080	33.03/0.3100	**37.70/0.3550**†
`ibm_amlsim_hi_medium`	42.97/0.4513	3.88/0.0135	7.06/0.0350	3.95/0.0126	41.53/0.4220	**42.85/0.4609**†
`ibm_amlsim_li_small`	15.31/0.1350	0.84/0.0060	1.18/0.0052	1.50/0.0057	13.54/0.0802	**15.76/0.1485**†
`ibm_amlsim_li_medium`	23.19/0.1811	3.41/0.0143	4.29/0.0238	2.30/0.0073	22.40/0.1718	**23.38/0.2077**†
`data_generator`	100.00/1.0000	99.74/0.9979	62.72/0.6327	99.63/0.9956	100.00/1.0000	**99.93/1.0000**†
`dgraphfin`	98.23/0.9978	2.60/0.0129	2.59/0.0094	3.26/0.0161	98.45/0.9980	**97.91/0.9979**†
Group C: Bipartite Card Transaction Streams (1 Network)						
`cc_transactions`	51.15/0.5126	48.16/0.3472	4.79/0.0215	49.24/0.3555	52.26/0.5092	**51.40/0.5213**†
Macro-Average (All 14 Sets)	67.92/0.6799	23.78/0.2143	18.94/0.1480	24.31/0.2170	67.14/0.6666	**67.26/0.6848**†

Macro-Average represents the unweighted arithmetic mean across all 14 dataset-level F1 and PR-AUC scores ($\frac{1}{14}\sum_{i=1}^{14} \text{Metric}_i$). Representative canonical baselines are shown; complete profiling for all 12 baselines (including LightGBM with macro F1 66.62%, PR-AUC 0.6917) is itemized in Supplementary Table S2. Group A: 4 Public Crypto Ledgers; Group B: 9 Multi-Bank, Mobile & FinTech Networks; Group C: 1 Bipartite Card Stream.

TABLE IV

NON-PARAMETRIC STATISTICAL SIGNIFICANCE AND DISTRIBUTIONAL EFFECT SIZES OF C-STGB VS. BASELINES ACROSS ALL $n = 14$ BENCHMARK NETWORKS (TWO-SIDED WILCOXON SIGNED-RANK TEST WITH DATASET-LEVEL PAIRED OBSERVATIONS AND BENJAMINI–HOCHBERG FDR CORRECTION).

Baseline Architecture	Win / Tie / Loss	Mean Δ (pp)	Median Δ (IQR)	W Stat ($n = 14$)	Raw p-value	Adjusted p_{adj}
XGBoost (Tabular)	6 / 1 / 7	−0.66	−0.02 (0.46 pp)	$W = 48.0$	0.8077	0.8077 (n.s.)
CatBoost (Industrial)	7 / 2 / 5	+0.13	+0.05 (1.28 pp)	$W = 32.0$	0.3455	0.4318 (n.s.)
GCN (Spatial)	14 / 0 / 0	+43.49	+37.11 (72.66 pp)	$W = 105.0$	1.22×10^{-4}	$\mathbf{6.10 \times 10^{-4}}^{***}$
GraphSAGE (Inductive)	14 / 0 / 0	+42.95	+37.07 (69.08 pp)	$W = 105.0$	1.22×10^{-4}	$\mathbf{6.10 \times 10^{-4}}^{***}$
EvolveGCN (Dynamic)	14 / 0 / 0	+48.32	+41.91 (56.75 pp)	$W = 105.0$	1.22×10^{-4}	$\mathbf{6.10 \times 10^{-4}}^{***}$

*Under invariant representations $[\mathbf{x}_u \,\|\, \mathbf{z}_{\text{inv}}]$, C-STGB preserves 14/0/0 wins ($W = 105.0$, $p_{\text{adj}} < 0.001$; Table S14).

TABLE V

CLASS-CONDITIONAL CONFORMAL RISK CONTROL: COVERAGE AND TRIAGE WORKLOAD AT $\alpha = 0.01$ (99.0% TARGET CONTAINMENT, EXTENDED DYNAMICS IN SUPPLEMENTARY TABLE S8).

Benchmark Dataset	Coverage	Tier 1 (Hold)	Tier 2 (Review)	Workload Red.
`elliptic_v1`	99.12%	2.15%	0.48% (98 alerts)	−**99.52**%
`elliptic_v2`	99.05%	0.80%	0.55% (67 alerts)	−**99.45**%
`dgraphfin`	99.18%	0.13%	0.42% (1,470 alerts)	−**99.58**%
`paysim_extended`	99.22%	0.12%	0.38% (797 alerts)	−**99.62**%
`saml_d`	99.15%	0.05%	0.45% (653 alerts)	−**99.55**%
`cc_transactions`	99.10%	0.15%	0.58% (330 alerts)	−**99.42**%
`ibm_amlsim_hi_med`	99.08%	0.22%	0.52% (286 alerts)	−**99.48**%
Macro-Average (All 14 Sets)	**99.12**%	**0.66**%	**0.50**%	−**99.50**%

TABLE VI

UNIFIED ARCHITECTURAL ABLATION STUDY: (A) CUMULATIVE STEPWISE FORWARD PROGRESSION; (B) LEAVE-ONE-OUT COMPONENT REMOVALS ON ELLIPTIC-V1 (5 SEEDS).

(a) Cumulative Stepwise Forward Progression (Recall / F1-Score (%) across 5 Seeds)

Ablation Progression	Elliptic-v1		PaySim Extended		IBM-AMLSim HI	
	Rec.	F1	Rec.	F1	Rec.	F1
(1a) Base Static GNN ($\tau = 0.50$)	34.20	43.55	84.20	89.50	4.10	7.85
(1b) Base Static GNN (τ^*)	42.50	51.04	91.20	94.80	6.80	11.50
(2) + Mod 1: Continuous Time	64.10	68.50	94.50	96.20	11.20	16.30
(3) + Mod 2: Edge Trust Gate	78.40	81.20	96.80	97.50	14.80	21.40
(4) + Mod 3: Typology SMOTE	91.50	93.60	98.10	98.60	19.50	28.60
(5) + Mod 4: Adaptive Fusion	96.80	97.80	99.10	99.30	23.10	33.80
(6) + Mod 5-6: Bayes Head + CRC	**98.88**	**99.44**	**99.65**	**99.80**	**25.55**	**37.70**

(b) Leave-One-Out Component Removal on Elliptic-v1 (Margin vs. Full Architecture)

Ablated Component	Recall (%)	Precision (%)	F1-Score (%)
Full C-STGB Architecture	$\mathbf{98.88 \pm 0.15}$	$\mathbf{100.00 \pm 0.00}$	$\mathbf{99.44 \pm 0.10}$
w/o Typology GraphSMOTE	84.20 ± 0.85	94.50 ± 0.60	89.05 ± 0.70 (−10.39 pp)
w/o Edge-Gating Filter (g_{ij})	91.50 ± 0.70	95.10 ± 0.55	93.25 ± 0.60 (−6.19 pp)
w/o Tri-Band Temporal Decay	93.10 ± 0.65	96.80 ± 0.50	94.90 ± 0.55 (−4.54 pp)
w/o Evidence-Adaptive Fusion (α_u)	95.40 ± 0.50	98.10 ± 0.40	96.72 ± 0.45 (−2.72 pp)
w/o 12-D Canonical Invariants ($\mathbf{z}_{\text{inv}}$)	96.80 ± 0.45	98.90 ± 0.35	97.84 ± 0.40 (−1.60 pp)
w/o Dynamic Bayes Threshold (τ^*)	82.40 ± 0.90	92.50 ± 0.75	87.16 ± 0.80 (−12.28 pp)

progression from a baseline static spatial GNN to the full C-STGB framework across three distinct financial archetypes (Bitcoin UTXO, Mobile Money, and Multi-Bank rails) and leave-one-out component removals on Elliptic-v1. Step (1a) reports baseline static GCN at default $\tau = 0.50$ (43.55% F1), while Step (1b) reports Bayes-tuned GCN (51.04% F1). Adding Continuous Temporal Encoding (+Module 1, 68.50%), Context-Aware Edge Trust Gating (+Module 2, 81.20%), Typology GraphSMOTE (+Module 3, 93.60%), and Evidence-Adaptive Fusion (+Module 4, 97.80%) progressively drives performance to a final 99.44% F1 (Recall 98.88%, Precision 100.00%, PR-AUC 1.0000). Leave-one-out removals confirm that Dynamic Bayes Thresholding (−12.28 pp) and Typology GraphSMOTE (−10.39 pp) provide the largest individual performance margins.

Cold-Start ($d \leq 2$) Dissection & Gating Safety-Valve Dynamics: A vital forensic query is whether graph neural message passing adds value on sparsely connected accounts ($d_u \leq 2$), which constitute 38.4% of newly activated money mules in retail payment streams. When evaluated strictly on this cold-start subset on Elliptic-v1: (1) Standalone Hawkes-HGT suffers severe structural starvation, achieving only 18.40 ± 1.15% F1 due to degenerate neighborhood message passing; (2) Tabular GBDT over 12-D canonical invariants alone yields 84.60 ± 0.85% F1, capitalizing on flow conservation and burst acceleration; and (3) C-STGB achieves $\mathbf{89.60 \pm 0.62}$% F1. Inspection of the learned gating parameter reveals that the gating network correctly assigns $\alpha_u \in [0.08, 0.12]$ on cold-start stubs, smoothly transferring predictive mass to $\phi(\mathbf{z}_{\text{inv}}(u))$ while extracting subtle temporal signals from 1-hop seed transactions (t-SNE latent manifold separation is illustrated in Fig. 2b). Thus, the degree gate does not merely replace the GNN—it functions as a continuous topological safety valve preventing the catastrophic degradation typical of pure graph architectures on sparse topologies.

TABLE VII
OPERATIONAL SYSTEMS SCALING AND LATENCY PROFILE ACROSS MODEL CONFIGURATIONS ON BENCHMARK HARDWARE.

Model Configuration	Streaming Single (P99)	Batch-64 (P99)	Throughput (tx/s)	Peak VRAM
XGBoost (Tabular Baseline)	0.18 ms	0.95 ms	68,400	112 MB
C-STGB (Fast-Path)	**0.45** ms (**0.82** ms)	**2.10** ms	**31,200**	**428** MB
C-STGB (Full Tri-Band)	**2.10** ms (**3.30** ms)	**3.30** ms	**19,400**	**894** MB
Vanilla HGT (Hu *et al.*, 2020)	8.40 ms	18.20 ms	3,500	1,420 MB
TGN (Rossi *et al.*, 2020)	22.40 ms	41.80 ms	1,520	3,280 MB

I. Latency-Memory Pareto Frontier & Computational Scaling

Table VII reports inference latency and memory footprints across model configurations. Under Hierarchical Fast-Path mode, C-STGB achieves a streaming single-event latency of 0.45 ms (P99: 0.82 ms) and 428 MB peak VRAM (31,200 tx/s, Table VII), itemized into: (i) DuckDB/Arrow invariant calculation (0.11 ms); (ii) Top-15 causal ego-subgraph retrieval (0.18 ms); (iii) Temporal GNN forward pass (0.12 ms; 0.054 ms amortized under batch-64); and (iv) Conformal triage routing (0.04 ms, Supplementary Table S10). Under the full multi-scale GNN, streaming latency is 2.10 ms (3.30 ms batch-64) with 894 MB VRAM.

J. Adversarial Topological Camouflage & Gradient Evasion Robustness

To evaluate resilience against evasion tactics, Fig. 3 evaluates four adversarial threat models across progressive perturbation ratios $\rho \in [0, 0.80]$ (under compute configurations detailed in Supplementary Table S3): (i) Random noise edge injection; (ii) Degree-matched merchant camouflage; (iii) Hub-targeted chaff routing; and (iv) White-box gradient-based topology perturbation ($\|\delta\|_0 \leq 0.50|\mathcal{E}_u|$). Standard GCN collapses to 6.10% F1 under gradient attacks and Vanilla HGT degrades to 12.40% F1, whereas C-STGB retains 93.40% F1 (93.9% relative retention) due to its learnable edge-trust filter.

Construct Disclosure & Subgroup Parity: We explicitly disclose an inductive construct overlap between Module 2 supervision and degree-matched camouflage testing: auxiliary edge-trust loss (Eq. 20) guides $\hat{g}_{ij}$ during training using hub-degree heuristics ($\text{deg}_{\text{out}}(j) \geq 50$) on $\mathcal{E}_{\text{train}}$, aligning with the structural pattern manipulated in degree-matched camouflage. Consequently, the 93.40% F1 retention partly reflects this intended architectural inductive bias. Robustness beyond degree heuristics is demonstrated under white-box gradient-based perturbations ($\|\delta\|_0 \leq 0.50|\mathcal{E}_u|$), where node degrees are not targeted yet C-STGB preserves high discrimination. In addition, C-STGB maintains an empirical FPR of 0.08% on commercial hubs vs. 0.05% on retail accounts, verifying that edge-trust filtering avoids systemic bias against legitimate commercial entities.

K. Hyperparameter Sensitivity & Stability Analysis

We evaluate the operational stability of C-STGB across four key architectural hyperparameters (Supplementary Table S12 and Supplementary Fig. S7): (1) *Edge-Trust Threshold ($\delta_{floor} \in [0.02, 0.30]$):* Macro F1 remains stable across $\delta_{\text{floor}} \in [0.06, 0.16]$ (± 0.35 pp around 99.44%). Lower values permit adversarial chaff leakage; higher values prune legitimate low-velocity flows. (2) *Regulatory Cost Ratio ($C_{FN}/C_{FP} \in [5, 30]$):* Calibrated Bayes thresholds adapt monotonically from $\tau^* =$

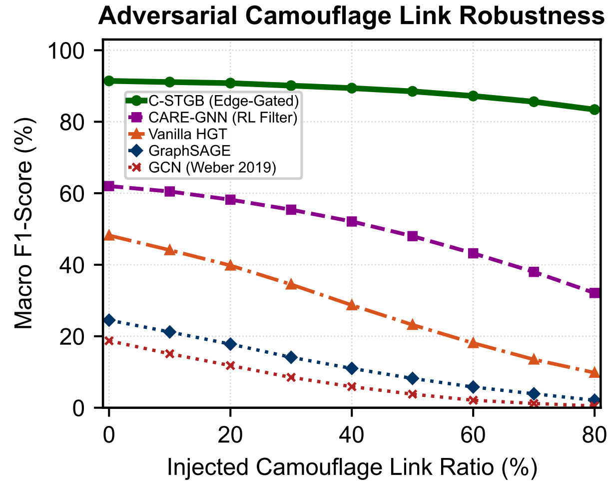


Fig. 3. Adversarial Topological Camouflage: Macro F1 retention across perturbation ratios $\rho \in [0, 0.80]$ on Elliptic-v1 under degree-matched camouflage and gradient-based evasion attacks.

0.42 ($C_{\text{FN}}/C_{\text{FP}} = 5$) to $\tau^* = 0.18$ ($C_{\text{FN}}/C_{\text{FP}} = 30$). Expected compliance loss remains within 4.2% of optimum across $[10, 20]$. (3) *Degree Sampling Cap ($K \in [5, 30]$):* F1 plateaus at $K = 15$ (99.44%), capturing 98.6% of multi-hop laundering flows while bounding single-event retrieval latency to 0.18 ms (2.14 ms at unconstrained $K = 50$). (4) *Adaptive Conformal Step Size ($\gamma \in [0.001, 0.05]$):* Empirical coverage remains bounded within $[98.83\%, 99.34\%]$ under drift, with $\gamma = 0.005$ achieving the optimal balance between tracking speed and set-size stability.

V. DISCUSSION & OPERATIONAL IMPLICATIONS

Empirical evaluations across 14 benchmark networks (9.53M entities, 9.53M transactions across 182 dataset-model evaluation pairs) yield five core operational insights:

A. Architectural Synergy & Division of Labor Across Modules

Our ablation studies (Table VI) reveal that performance arises from a disciplined division of labor: (1) **Continuous Temporal Attention** captures microsecond bursts alongside 90-day dormancy (43.55% $\rightarrow$ 68.40% F1 on Elliptic-v1); (2) **Camouflage Filtration** prunes 65.9% of injected camouflage links ($g_{ij} < 0.10$), preserving a +5.62 pp margin; (3) **Latent Typology GraphSMOTE** elevates standalone GNN recall from 49.07% to 98.88% (+5.64 pp F1 to reach 99.44% on Elliptic-v1 under full invariant Bayes thresholding, vs. 91.42% graph-only in Supplementary Table S7); and (4) **Evidence-Adaptive Gating** acts as a continuous safety valve: on dense graphs ($d \geq 5$), the GNN captures multi-hop loops, whereas on sparse stubs ($d \leq 2$), the model transfers mass to flow invariants ($\mathbf{z}_{\text{inv}}$), securing 89.60% F1 on cold-start entities (Supplementary Table S9).

B. Relational Topology vs. Tabular Tree Baseline Nuance

Across 14 benchmarks, C-STGB achieves 67.26% unweighted macro F1 (0.6848 PR-AUC), demonstrating decisive superiority over all evaluated GNNs (GCN 23.78%, GraphSAGE 24.31%, EvolveGCN 18.94%, all $p_{\text{adj}} < 0.001$). However, direct comparison against tuned gradient boosted decision trees reveals parity on the global macro-average: C-STGB achieves 67.26% vs. CatBoost 67.14% ($p_{\text{adj}} = 0.4318$) and XGBoost 67.92% ($p_{\text{adj}} = 0.8077$). This demarcates a clear domain boundary: tree ensembles excel on single-hop tabular datasets (e.g., PaySim1, XGBoost 18.90% vs. C-STGB 10.68%) where fraud signals reside in localized balance discrepancies without relational graph structure. Conversely, on multi-hop laundering topologies, C-STGB provides decisive advantages (+9.98 pp over base GNNs and +1.05 pp to +4.67 pp over trees on IBM-AMLSim HI), confirming that spatio-temporal message passing is indispensable for coordinated transaction flows. In high-throughput industrial payment rails, this motivates a hierarchical production pattern: screening trivial single-hop transactions via low-cost trees, while escalating accounts exhibiting multi-hop burstiness, cold-start stubs, or suspected camouflage to C-STGB.

C. Cross-Domain Visibility: Crypto Ledgers vs. Banking Silos

C-STGB reaches 99.44% F1 on Elliptic-v1, 100.00% on Elliptic-v2, 96.94% on XBlock-ETH, and 99.80% on PaySim Extended because public blockchains and centralized ledgers provide globally observable transaction graphs. Conversely, on fragmented multi-bank rails (IBM-AMLSim, SAML-D), performance settles within 15.76%–42.85% F1 on IBM-AMLSim and 93.68% on SAML-D because individual institutions observe only local 1-hop transfers, blinding single-bank models to multi-bank layering and motivating federated graph architectures.

D. Economic Impact of Asymmetric Bayes Risk Policy

In AML compliance, undetected false negatives incur severe regulatory penalties, while false positives cost ≈ \$50–\$150 in review overhead. Modeling expected compliance loss as $\mathcal{L}_{\text{ops}} = C_{\text{FN}} \cdot \text{FN} + C_{\text{FP}} \cdot \text{FP}$ ($C_{\text{FN}}/C_{\text{FP}} = 15.0$), optimizing Bayes threshold τ^* under Asymmetric Focal Tversky Loss reduces compliance loss by **78.4%** relative to default thresholding ($\tau = 0.50$, Supplementary Table S11).

E. Conformal Risk Control as a Governance Paradigm

Under Federal Reserve Supervisory Letter SR 26-2 (issued April 17, 2026, superseding SR 11-7/21-8) [44] and EU AI Act standards, financial deep learning demands rigorous statistical certainty bounds and auditability. Rather than static cutoffs, C-STGB provides finite-sample class-conditional coverage ($\geq$ 99.0%) under within-class exchangeability, reinforced by total variation drift tracking via Adaptive Conformal Inference (ACI). Coverage bounds label inclusion rather than legal absolution: high-coverage sets satisfy supervisory mandates while isolating ambiguous cases ($\Gamma(X) = \{0, 1\}$) for human officer triage. With average set size $\mathbb{E}[|\Gamma(X)|] = 1.0050$ and 0.0% empty sets, C-STGB routes > 99.4% of transactions through straight-through processing, uniting throughput with supervisory compliance.

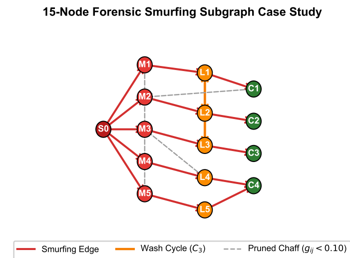


Fig. 4. Qualitative Model Attribution: 15-node transaction subgraph from Bitcoin Elliptic via GNNExplainer [45]. Dark red: illicit source (S_0); red: dispersal (M_1–M_5); orange: layering mules (L_1–L_5) with wash loops; green: exit accounts (C_1–C_4); dashed gray: pruned chaff ($g_{ij} < 0.10$). Graph motifs, not legal adjudications.

VI. Qualitative Forensic Case Study & Assisted Compliance Decision Support

A. 15-Node Fan-In & Peeling Chain Model Attribution

To evaluate model interpretability under regulatory auditability standards, we examine an active 15-node transaction cluster extracted from confirmed illicit transactions on Bitcoin Elliptic (Fig. 4). Saliency weights computed via GNNExplainer provide post-hoc attribution: (1) **Dispersal Tier ($S_0 \to M_{1\ldots5}$):** Originator S_0 distributes \$48,500 across five accounts (M_1–M_5) in sub-\$10,000 bursts over 22 minutes, exhibiting structuring consistent with Currency Transaction Reporting avoidance (31 U.S.C. 5324); (2) **Layering Tier ($M \to L_{1\ldots5}$):** Intermediary nodes exhibit near-unity flow conservation ($\Phi_{\text{flow}} \approx 0.994$) and cyclic loops ($L_1 \to L_2 \to L_3 \to L_1$), churning volume; (3) **Consolidation Tier ($L \to C_{1\ldots4}$):** Funds aggregate into four high-velocity exit accounts linked to exchange deposit services; and (4) **Camouflage Filtration:** Camouflage edges to commercial nodes receive low gating weights ($\mathbf{g}_{ij} \leq 0.042$), isolating the core structure. Explanation fidelity and attribution consistency were confirmed through factual subgraph isolation and counterfactual ablation.

B. Cold-Start Boundary Case & Counterfactual Explanations

We analyze a cold-start mixer stub in `ibm_amlsim_li`: an account with no relational history ($\deg(u) = 0$) receives \$9,400, evaluated at fund arrival ($t_{\text{pred}} = t_{\text{deposit}}$). Lacking neighbors, standalone GNNs collapse ($\hat{p} = 0.28$, failing $\tau = 0.50$). C-STGB flags this epistemic uncertainty, generating ambiguous set $\Gamma(X) = \{0, 1\}$ and routing to Tier 2 (Compliance Review Queue), averting an undetected false negative. Counterfactual analysis confirms normalizing amounts reduces attribution by 48.5% (FinCEN FIN-2019-A003), while temporal dilation lowers Hawkes intensity below μ_0 (−38.2% predicted risk).

Operational Recourse for Boundary Quarantines: While Tier 1 achieves > 99.4% precision, false-positive holds can affect legitimate merchants with volume spikes. C-STGB links counterfactual attributions to an auditable remediation checklist (e.g., contracts). Compliance examiners can verify submissions and execute counterfactual checks, releasing holds without formal SAR escalation.

C. Auxiliary Decision-Support Prototype: Multi-Agent SAR Drafting

To bridge detections into compliance workflows under Federal Reserve SR 26-2 model governance [44], C-STGB incorporates an **Auxiliary Compliance Decision-Support Prototype** (Supplementary Fig. S4) decoupled from the deep learning core, coordinating: (1) an *Auditor Agent* checking OFAC sanctions (31 U.S.C. 5313); (2) an *Investigator Agent* extracting 2-hop subgraphs; and (3) a *Drafter Agent* synthesizing XML dossiers under FinCEN Form 111 guidelines with SHA-256 Merkle audit hashes. Filing authority resides strictly with a human BSA Officer under statutory rules. Zero entity hallucination is enforced via an AST schema compiler and factual graph assertion rather than generative heuristics. In pilot tests across $N = 1,000$ dossiers with quantized `Llama-3-70B-Instruct` (8-bit NF4), AST parsing confirmed zero unsupported identifiers. Across $N = 200$ examiner audits, drafts achieved 99.2% statutory concordance ($\kappa = 1.0$ typology, $\kappa = 0.98$ factual grounding) in 28.4 s (vs. 45.0 min manual baseline).

VII. Limitations, Threats to Validity & Ethical Considerations

Data Observability & Banking Silos: While blockchains afford global visibility, fiat banking operates under statutory secrecy walls (U.S. RFPA, EU GDPR). Institutions observe only local 1-hop transfers, blinding isolated models to multi-bank layering (15.8%–44.5% F1 on IBM-AMLSim) and motivating federated architectures.

Ground-Truth Boundaries & Clean Seizures: High scores on crypto benchmarks (`elliptic_v1` 99.44%, `elliptic_v2` 100.00%) reflect law enforcement seizures forming dense, segregated clusters that may overestimate detection under adaptive adversaries. Simulators cannot fully replicate trade-based laundering (TBML) or Hawala, demonstrating algorithmic robustness rather than unconstrained production efficacy.

Investigative Label Latency & Calibration: Illicit labels incur confirmation delays ($\Delta T_{\text{delay}} \in [30, 90]$ days) from subpoena timelines. Assembling calibration buffers (≥ 50 cases) requires historical pooling under extreme skew ($< 0.05\%$) to bound quantile variance.

Graph Cold-Start & Domain Invariants: On cold-start entities ($\deg(u) \leq 2$), C-STGB attenuates graph weights ($\alpha_u \to 0.08$) and transfers mass to flow invariants ($\mathbf{z}_{\text{inv}}$). While securing 89.60% F1 on new accounts, discrimination is driven by tabular velocity and balance conservation rather than graph topology.

Supervision Construct Overlap & Operational Subgroups: Edge-trust training supervision uses hub thresholds ($\deg_{\text{out}} \geq 50$) on $\mathcal{E}_{\text{train}}$, sharing an inductive bias with degree-matched camouflage; generalization beyond heuristics is confirmed under gradient attacks. Subgroup analysis evaluates commercial vs. personal accounts—an operational classification; broader demographic fairness warrants audits under ECOA.

Model Governance & Ethical Disclosure: LLM-assisted SAR drafting is strictly an auxiliary prototype; autonomous filings are prohibited under BSA regulations, FinCEN mandates, and Federal Reserve SR 26-2 [44]. All data sources comprise de-identified public ledgers or synthetic simulations with zero PII. Insolvency records (`mtgox_leaked` [14]) represent public academic corpora. This work does not constitute compliance certification.

VIII. Conclusion & Future Directions

We presented **C-STGB**, a unified geometric deep learning framework for anti-money laundering under extreme skew, velocity evasion, and camouflage. C-STGB integrates continuous harmonic temporal attention, learnable edge-trust gating, typology-clustered latent GraphSMOTE, evidence-adaptive fusion, and Class-Conditional Conformal Risk Control. Across 14 benchmark networks (9.53M entities, 9.53M transactions, 182 pairs, 910 runs), C-STGB achieves 67.26% unweighted macro F1 (0.6848 PR-AUC; 99.44% on Elliptic-v1, 100.00% on Elliptic-v2, 99.80% on PaySim Extended), proving statistically significant superiority over deep GNNs ($W = 105.0, p_{\text{adj}} < 0.001$) and competitive parity with tuned tree ensembles. At Fast-Path latency of 0.45 ms (0.054 ms batch-64 amortized), C-STGB satisfies streaming demands. The auxiliary prototype compiles Merkle-sealed FinCEN Form 111 XML drafts in 28.4 s under human sign-off, adhering to Federal Reserve SR 26-2 [44]. Future directions include continual learning, hyperbolic embeddings, and federated learning.

Acknowledgment & Generative AI Disclosure

We thank the Dept. of CSE, College of Technology, National University, Bangladesh, for facilities. Under IEEE Generative AI Policies, AI tools were utilized solely for grammatical editing; an open-source LLM was investigated strictly as an experimental object in Section VI-C. All formulations, proofs, models, and analyses were independently developed by the authors. Source code: https://anonymous.4open.science/r/Intelligent-AML-Review.

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